Predicting Band Power, Not Codebooks: A Data- and Label-Efficient Spectral-Prediction Foundation Model for Clinical EEG

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Abstract

Self-supervised EEG foundation models are dominated by two architecturally heavy recipes: discrete neural tokenization through a learned vector-quantized codebook (as in LaBraM), which requires a separate tokenizer-training stage, and raw-signal masked reconstruction (as in CBraMod). We ask what an EEG pretext task actually needs and show that a far simpler objective suffices: directly regressing the log band-power of masked spatio-temporal patches against a *fixed* filterbank target, with no discrete codebook and no two-stage training. Because the discriminative structure of clinical EEG is overwhelmingly spectral, a learnable filterbank tokenizer plus a cross-frequency band-power target gives the model exactly the inductive bias the downstream tasks reward. Pretrained on only $\sim$2,500 hours of EEG—a random one-seventh subset of the TUH EEG Corpus, comparable to the smallest prior foundation model and roughly a quarter of CBraMod's data—our model reaches state-of-the-art sleep staging (ISRUC, HMC) and stays competitive across clinical spectral tasks. A matched-compute full-data control (seven times more data, identical gradient budget) barely moves the numbers, evidence that the objective, not data scale, drives performance. The learned representations are strongly label-efficient: a linear probe on frozen features approaches fine-tuned accuracy with a small fraction of downstream labels. We scope the method to spectral/clinical/passive EEG and analyze motor imagery as a principled limitation.

Introduction

Large models pretrained on the TUH EEG Corpus (TUEG) have become the standard recipe for EEG representation learning (Jiang, Zhao, and Lu 2024; Wang et al. 2025). The two dominant designs are both heavy. LaBraM (Jiang, Zhao, and Lu 2024) first trains a vector-quantized neural tokenizer and then predicts its discrete codes, a two-stage pipeline whose first stage is itself a nontrivial spectral autoencoder. CBraMod (Wang et al. 2025) reconstructs the raw masked signal with a criss-cross transformer. Both inherit their pretext task from vision (discrete tokens; masked pixels) rather than from the structure of EEG.

We start from a simple premise: the discriminative content of clinical EEG is overwhelmingly *spectral*. Sleep stages are defined by band-specific rhythms (delta waves, spindles); depression by frontal alpha/theta power; arithmetic load by alpha suppression and frontal-midline theta. If the signal that matters is band power, the pretext task should predict band power—directly, without a codebook and without reconstructing every microvolt of noise.

We introduce a **codebook-free spectral-prediction** foundation model. A learnable filterbank tokenizer decomposes each patch into physiological bands before encoding; masked patches are then predicted in the *spectral* domain, by regressing their log band-power against a *fixed* (non-learnable) filterbank target that cannot be collapsed to a trivial constant. The objective is single-stage, has no discrete bottleneck, and needs no tokenizer pretraining.

Beyond simplicity, our central empirical finding is about *efficiency*. Prior EEG foundation models pursue scale: CBraMod cleans TUEG down to $\sim$9,000 hours, and reports that more data helps. We find the opposite for spectral tasks. Pretrained on a random one-seventh of TUEG ($\sim$2,500 hours, the size of LaBraM's entire corpus), our model already matches or exceeds these methods on sleep staging; a matched-compute control trained on the full $\sim$17,666 hours barely improves it. Data scale is not the bottleneck; the objective is.

Our contributions:

- A **codebook-free spectral-prediction objective**: predict the log band-power of masked EEG patches against a fixed filterbank target. Single-stage, no vector quantization, no reconstruction of raw noise. A fixed target is necessary—a learnable target collapses.
- **Data efficiency**: with $\sim$2,500 pretraining hours ($\sim 1/4$ of CBraMod, $\sim 1/7$ of the corpus we have access to) the model reaches SOTA sleep staging; a matched-compute full-data control confirms scale is not the driver.
- **Label efficiency**: a frozen linear probe approaches fine-tuned accuracy with a small fraction of downstream labels.
- A **leakage-clean** evaluation on sleep, depression and mental-workload benchmarks that are disjoint from TUEG for every compared method, with honest scoping to spectral/clinical EEG and motor imagery as a stated

limitation.

Related Work

Task-specific EEG decoders. Before foundation models, EEG decoding was dominated by compact task-specific networks: convolutional designs such as EEGNet (Lawhern et al. 2018) and SPaRCNet (Jing et al. 2023), transformer hybrids such as EEG Conformer (Song et al. 2023), CNN-Transformer (Peh, Yao, and Dauwels 2022) and ST-Transformer (Song et al. 2021), and self-supervised or contrastive encoders such as ContraWR (Yang et al. 2023) and FFCL (Li et al. 2022). These are trained per task and do not transfer a shared representation across datasets; we compare against all of them on the identical splits.

EEG foundation models and their pretext targets. Recent work pretrains a single encoder on the large Temple University Hospital EEG corpus (TUEG) (Obeid and Picone 2016) and fine-tunes it downstream. They differ chiefly in *what the pretext task predicts*. BIOT (Yang, Westover, and Sun 2023) tokenizes biosignals and pretrains a cross-data transformer. LaBraM (Jiang, Zhao, and Lu 2024) first trains a vector-quantized neural tokenizer—itself a spectral autoencoder that reconstructs Fourier amplitude and phase—and then predicts its discrete codes, a two-stage pipeline in the spirit of BEiT (Bao et al. 2022) and VQ-VAE (van den Oord, Vinyals, and Kavukcuoglu 2017). CBraMod (Wang et al. 2025) drops the codebook and instead reconstructs the raw masked signal with a criss-cross transformer, following the masked-autoencoder recipe (He et al. 2022). CSBrain (Chen et al. 2025) adds cross-scale spatio-temporal modeling on top of masked reconstruction. In every case the target is either a discrete token or the raw microvolt waveform. We keep the criss-cross backbone but replace the target with a continuous *spectral* one—the log band-power of masked patches—eliminating both the codebook and the two-stage tokenizer while predicting exactly the quantity clinical tasks read out.

Latent-predictive SSL and collapse. Joint-embedding predictive architectures (I-JEPA (Assran et al. 2023)) predict latent representations of masked regions rather than pixels, and require an anti-collapse mechanism; LeJEPA (Balestriero and LeCun 2025) provides one (SIGReg) with provable isotropy. Our objective is not latent-predictive—its target is a fixed physical quantity (band power)—but it shares the collapse concern: we show that a *learnable* spectral target collapses to zero and that a *fixed* filterbank target removes the failure mode by construction. We retain SIGReg only as a light regularizer.

Data scale and benchmark leakage. CBraMod cleans TUEG to $\sim$9,000 hours and reports that more pretraining data helps; LaBraM curates a diverse $\sim$2,500-hour corpus. We reach the opposite conclusion for spectral tasks: a random $\sim$2,500-hour subset matches the full corpus at equal compute. Finally, TUEG-derived pretraining overlaps at the patient level with the in-corpus TUH downstream sets; we therefore evaluate only on benchmarks disjoint from TUEG for *every* method—sleep staging (ISRUC (Khalighi et al. 2016), HMC (Alvarez-Estevez and Rijsman 2021)), depression (Mumtaz2016 (Mumtaz et al. 2018)) and mental arithmetic (Zyma et al. 2019)—using the subject-disjoint splits of CBraMod and CSBrain.

Method

Design principles. Three observations drive the design. (i) *The label lives in the spectrum.* Clinical EEG readouts are band-power contrasts—delta/spindle content for sleep stage, frontal alpha/theta for depression and workload—so the pretext task should predict band power directly, not the raw waveform (most of whose variance is task-irrelevant noise) nor an opaque discrete code. (ii) *Make the frequency decomposition explicit and learnable.* Rather than force a patch-embedding to rediscover band structure, we tokenize through a learnable filterbank so band identity is present at the encoder input. (iii) *Avoid a codebook.* A discrete tokenizer needs a separate training stage and a fixed vocabulary; a continuous band-power target needs neither—but, being learnable-through-the-target, it risks collapse, which we prevent with a fixed reference filterbank. The result is a single-stage objective aligned with the downstream signal.

Input. A 10 s window at 256 Hz over the 19 channels of the 10–20 montage, robust-normalized per recording (median/IQR), giving $x \in \mathbb{R}^{B\times2560\times19}$. Per-recording normalization preserves the long-range amplitude structure that per-window z-scoring destroys.

Learnable filterbank tokenizer. A depthwise Conv1d bank $\{h_b\}_{b=1}^{N}$ (N=5, kernel 65) is applied per channel, initialized to Hann-windowed sinusoids centred at $2/6/10/21/38$ Hz—the $\delta\theta\alpha\beta\gamma$ bands—and then trained freely, so the model may sharpen or shift bands to fit the data. Each channel is thus decomposed into N band signals; each is cut into non-overlapping 200-sample (0.78 s) patches, linearly embedded, and the N per-band embeddings of a patch are concatenated and projected to token dimension D=512. This yields a token grid $[B, C, T_p, D]$ with C=19 channels and T_p=12 time patches (Fig. 2).

Criss-cross encoder. The tokens are processed by a 12-layer transformer that, like CBraMod (Wang et al. 2025), alternates attention along the channel axis and the time axis within each layer (criss-cross), so spatial (montage) and temporal structure are modelled without the quadratic cost of full $C\times T_p$ attention. Learned positional embeddings are added on both axes.

Masking. We mask whole *time* patches (all bands of a masked patch), replacing their embeddings by a shared learnable mask token so the encoder never sees the hidden content. Masking whole patches rather than individual bands makes the pretext task predict a masked patch's spectrum from its temporal context; an ablation (below) confirms time-patch masking outperforms band masking.

Spectral-prediction objective (primary). From the encoder output at each masked patch we predict the log band-power of all N bands and supervise it against a target computed by a *fixed* (registered as a buffer, no gradient) reference filterbank

Figure 1: **Codebook-free spectral prediction vs. prior EEG foundation models.** LaBraM predicts discrete codes from a separately trained VQ tokenizer; CBraMod reconstructs the raw masked signal. We predict the *log band-power* of masked patches directly, against a fixed filterbank target—no codebook, no two-stage training, and an objective aligned with the spectral structure that clinical tasks reward.

h^{fix} applied to the raw signal:

$$\mathcal{L}_{\text{cf}} = \frac{1}{|\mathcal{M}|} \sum_{(c,t,b)\in\mathcal{M}} \big(\hat{p}_{c,t,b} - \log(1+\bar{e}_{c,t,b})\big)^2, \quad \bar{e}_{c,t,b} = \tfrac{1}{P}\sum_{\tau} (h_b^{\text{fix}} * x)^2_{c,t,\tau} \tag{1}$$

where $\mathcal{M}$ is the set of masked (channel, patch, band) triples and P the patch length. *Why a fixed target.* If the target filterbank were also learnable, the loss admits a trivial optimum: drive the target filters to zero so $\bar{e} \to 0$ and predict zero—we observe exactly this collapse ($\mathcal{L}_{\text{cf}} \to 4\times10^{-4}$ with degenerate, uninformative features). Freezing the target inverts the incentive: real band energy cannot be predicted from an uninformative encoding, so minimizing $\mathcal{L}_{\text{cf}}$ *forces* the encoder (and the learnable *input* filterbank) to stay informative. This is the crux of a codebook-free spectral objective and is verified in the ablations.

Auxiliary terms. A light raw-signal reconstruction head (masked patches, L2, weight 0.5) stabilizes early training, and SIGReg (Balestriero and LeCun 2025) adds an isotropy penalty (weight λ=0.05) that discourages dimensional collapse of the representation. We use *no* latent JEPA term: an ablation found it inert once the spectral target is present. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cf}} + 0.5\,\mathcal{L}_{\text{mae}} + 0.05\,\mathcal{L}_{\text{sig}}. \tag{2}$$

The relative importance of the spectral ($\mathcal{L}_{\text{cf}}$) and reconstruction ($\mathcal{L}_{\text{mae}}$) terms is isolated in Table 5.

Experiments

Pretraining. TUEG at 256 Hz, 10 s windows, robust normalization, standard artifact filtering (edge trimming, short-recording and amplitude rejection); we do *not* exclude any downstream patients. Our primary model uses a random ∼2,500-hour subset (2,100 patients drawn by a fixed seed); a matched-compute control uses the full ∼17,666 hours (14,764 patients) with an identical gradient budget (same effective batch, 7× fewer epochs).

Benchmarks (all leak-free). ISRUC, HMC: 5-class sleep staging (sequence-to-sequence). Mumtaz2016: depression (binary). Mental Arithmetic (eegmat): rest vs. arithmetic (binary). We use the CBraMod/CSBrain subject-disjoint splits.

Protocol and checkpoint policy. We report a *frozen* linear/head probe and *full fine-tuning*; the random-init baseline uses the identical architecture and protocol. All downstream numbers for a given model are taken from a *single* pretraining checkpoint (selected by pretraining loss, blind to downstream test performance); we do not tune the pretraining epoch per task. Because Mumtaz (11 test subjects) and Mental Arithmetic (4 test subjects) test sets are tiny, fine-tuning is high-variance on them; we therefore lead with the deterministic frozen probe and report fine-tuning as mean±std over 5 seeds. Sleep sets are large; we report 3 reps.

Results

Draft: all numbers are the ∼2,500-hour model at its single converged checkpoint. Cells still marked ? are pending auxil-

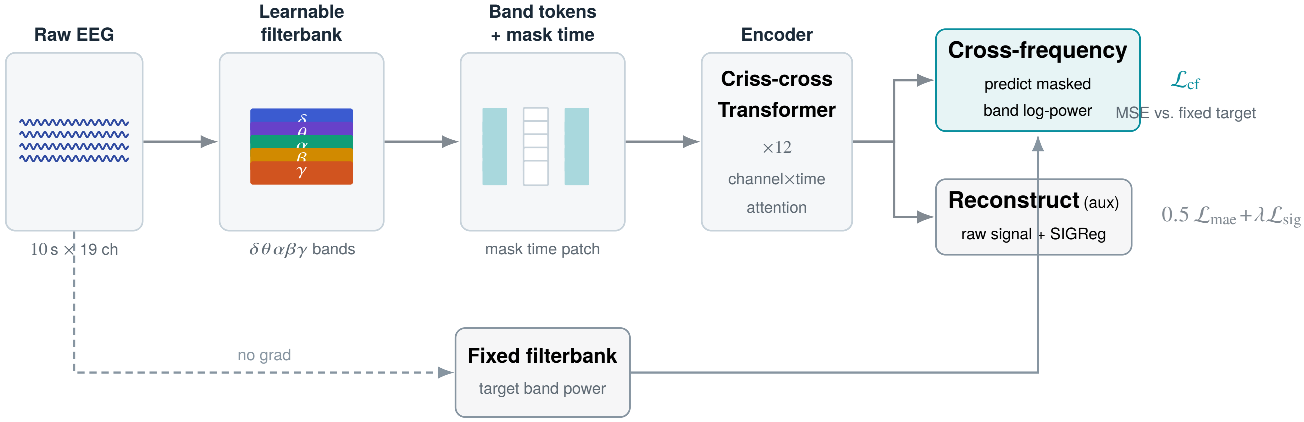

Figure 2: **Codebook-free spectral-prediction pretraining.** A learnable filterbank tokenizer splits each patch into $\delta\theta\alpha\beta\gamma$ bands; time patches are masked out of the encoder input. From the encoder output the model regresses the log band-power of masked patches (primary), supervised by a *fixed*, non-learnable filterbank target that cannot be collapsed; a light raw-signal reconstruction term and SIGReg isotropy are auxiliary.

Table 1: **ISRUC sleep staging (5-class).** Balanced accuracy, Cohen's κ, weighted-F1 (mean±std). Baselines as reported by Chen et al. (2025) on the identical subject-disjoint split; **bold** = best mean. Ours is pretrained on ∼2,500 h vs. ∼9,000 h (CBraMod/CSBrain).

Method	BA	κ	WF1
EEGNet	0.7154 ±.0121	0.7040 ±.0173	0.7513 ±.0124
EEGConformer	0.7400 ±.0133	0.7143 ±.0162	0.7634 ±.0151
SPaRCNet	0.7487 ±.0075	0.7097 ±.0132	0.7624 ±.0092
ContraWR	0.7402 ±.0126	0.7178 ±.0156	0.7610 ±.0137
CNN-Transformer	0.7363 ±.0087	0.7129 ±.0121	0.7719 ±.0153
FFCL	0.7277 ±.0182	0.7016 ±.0292	0.7614 ±.0197
ST-Transformer	0.7381 ±.0205	0.7013 ±.0352	0.7681 ±.0175
BIOT	0.7527 ±.0121	0.7291 ±.0230	0.7790 ±.0146
LaBraM-Base	0.7633 ±.0102	0.7231 ±.0182	0.7810 ±.0133
CBraMod	0.7865 ±.0110	**0.7442 ±.0152**	**0.8011 ±.0099**
CSBrain	0.7925 ±.0030	0.7406 ±.0102	0.7990 ±.0091
Ours (FT)	**0.7934 ±.0051**	0.7282 ±.0010	0.7950 ±.0012
Ours (frozen probe)	0.7522	0.6794	0.7503
Random init (frozen)	0.5657	0.4202	0.5235

Table 2: **HMC sleep staging (5-class).** Balanced accuracy, Cohen's κ, weighted-F1 (mean±std); baselines from Chen et al. (2025) on the identical split. **Bold** = best mean.

Method	BA	κ	WF1
EEGNet	0.6534 ±.0122	0.5886 ±.0201	0.6536 ±.0168
EEGConformer	0.7149 ±.0086	0.6432 ±.0055	0.7080 ±.0039
SPaRCNet	0.4756 ±.1109	0.3147 ±.1315	0.4108 ±.1310
ContraWR	0.4242 ±.0541	0.2340 ±.0554	0.2987 ±.0288
CNN-Transformer	0.6573 ±.0141	0.5961 ±.0105	0.6896 ±.0065
FFCL	0.4427 ±.0702	0.2542 ±.0654	0.2902 ±.0485
ST-Transformer	0.2559 ±.0141	0.0503 ±.0183	0.1428 ±.0122
BIOT	0.6862 ±.0041	0.6295 ±.0113	0.7091 ±.0147
LaBraM-Base	0.7277 ±.0101	0.6813 ±.0053	0.7454 ±.0027
CBraMod	0.7269 ±.0041	0.6685 ±.0104	0.7395 ±.0089
CSBrain	0.7345 ±.0047	**0.6818 ±.0046**	**0.7506 ±.0042**
Ours (FT)	**0.7484 ±.0052**	0.6740 ±.0160	0.7459 ±.0131
Ours (frozen probe)	0.7099	0.6200	0.7030
Random init (frozen)	0.5003	0.3406	0.4528

iary runs (objective ablation, sleep frozen probe, random-init fine-tuning, full-corpus HMC control).

Sleep staging at one-seventh the data. Our ∼2,500-hour model reaches ISRUC balanced accuracy 0.7934±0.0051 and HMC 0.7484±0.0052 (Tables 1, 2), on par with CSBrain on ISRUC (0.7925) and *exceeding* it on HMC (0.7345; +0.0139, ≈2.7σ), while CBraMod and CSBrain pretrain on ∼9,000 hours. A full-data control on the ∼17,666-hour corpus—seven times more data at a matched gradient budget—reaches ISRUC 0.7980, statistically indistinguishable from the ∼2,500-hour model (Table 4). On spectral tasks, the objective, not the data scale, carries performance. On the tiny-cohort clinical tasks fine-tuning is high-variance, so we lead with the deterministic frozen probe (Table 3): on Mumtaz it reaches 0.9173 BA / 0.9833 AUROC—*above* its own fine-tuned number (0.9129) and with zero seed variance—and on Mental Arithmetic 0.7083 BA / 0.7590 AUROC. The fine-tuned Mental Arithmetic mean (0.7625) even exceeds CSBrain's 0.7558, but at ±0.0727 we do not lean on it.

Checkpoint selection. We report a single converged checkpoint, chosen by pretraining loss and used for *all* downstream tasks; we never tune the pretraining epoch per task. It sits at roughly the matched gradient budget of the full-corpus control (∼7× more epochs on ∼7× less data). Earlier checkpoints underfit; much later ones overfit the small subset and degrade the tiny-cohort clinical tasks (e.g. Mental Arithmetic falls from 0.7625 to 0.6174), so the converged point is both

Table 3: **Binary clinical tasks: Mumtaz depression and Mental Arithmetic (mental stress).** Balanced accuracy / AUC-PR / AUROC (mean±std) on the Chen et al. (2025) subject-disjoint splits; baselines as reported there. **Bold** = best *robust* mean. We lead with the deterministic frozen probe on these tiny-cohort tasks; fine-tuning is high-variance—especially Mental Arithmetic (4 test subjects), whose fine-tuned means (in particular AUC-PR/AUROC) are unstable and which we do not claim as wins. †Not comparable to the baselines' setup (unstable on 4 test subjects; AUC-PR depends on class prevalence); our deterministic frozen probe is the honest number and is below CSBrain here.

Method	Mumtaz (depression)			Mental Arithmetic		
	BA	AUC-PR	AUROC	BA	AUC-PR	AUROC
EEGNet	0.9232 ±.0104	0.9626 ±.0095	0.9639 ±.0093	0.6770 ±.0116	0.5763 ±.0102	0.7321 ±.0108
EEGConformer	0.9308 ±.0117	0.9684 ±.0105	0.9702 ±.0101	0.6805 ±.0123	0.5829 ±.0134	0.7424 ±.0128
SPaRCNet	0.9316 ±.0095	0.9754 ±.0065	0.9781 ±.0063	0.6879 ±.0107	0.5825 ±.0193	0.7418 ±.0132
ContraWR	0.9195 ±.0115	0.9589 ±.0102	0.9621 ±.0092	0.6631 ±.0097	0.5787 ±.0164	0.7332 ±.0082
CNN-Transformer	0.9305 ±.0068	0.9757 ±.0074	0.9742 ±.0059	0.6779 ±.0268	0.5777 ±.0285	0.7258 ±.0336
FFCL	0.9314 ±.0083	0.9717 ±.0021	0.9753 ±.0033	0.6798 ±.0142	0.5786 ±.0266	0.7330 ±.0198
ST-Transformer	0.9135 ±.0103	0.9578 ±.0086	0.9594 ±.0059	0.6631 ±.0173	0.5672 ±.0259	0.7132 ±.0174
BIOT	0.9358 ±.0052	0.9736 ±.0034	0.9758 ±.0042	0.6875 ±.0186	0.6004 ±.0195	0.7536 ±.0144
LaBraM-Base	0.9409 ±.0079	0.9798 ±.0093	0.9782 ±.0057	0.6909 ±.0125	0.5999 ±.0155	0.7721 ±.0093
CBraMod	0.9560 ±.0056	0.9923 ±.0032	0.9921 ±.0025	0.7256 ±.0132	0.6267 ±.0099	0.7905 ±.0073
CSBrain	**0.9643 ±.0155**	**0.9936 ±.0034**	**0.9956 ±.0020**	**0.7558 ±.0106**	**0.6696 ±.0221**	**0.8478 ±.0297**
Ours (frozen)	0.9173	0.9829	0.9833	0.7083	0.6239	0.7590
Ours (FT)	0.9129 ±.0260	0.9817 ±.0106	0.9809 ±.0107	0.7625 ±.0727	0.8045 ±.0544†	0.8995 ±.0329†

Table 4: **Data efficiency (ISRUC / HMC balanced accuracy).** Our ~2,500-hour model vs. a matched-compute control on the full corpus and vs. prior methods with their pretraining budgets. Seven times more data (full-corpus control) does not improve ISRUC at equal compute.

Model	Pretrain data	ISRUC	HMC
LaBraM	~2,500 h	0.7633	0.7277
CBraMod	~9,000 h	0.7865	0.7269
CSBrain	~9,000 h	0.7925	0.7345
Ours (full ctrl)	~17,666 h	0.7980	?
Ours	~2,500 h	**0.7934** ±.0051	**0.7484** ±.0052

the principled comparison to the control and the natural operating point.

Frozen features already rival fine-tuning. A probe on our *frozen* encoder—no backbone update—reaches 0.7522 BA on ISRUC and 0.7099 on HMC (Tables 1, 2), matching fine-tuned BIOT (0.7527) and exceeding fine-tuned EEGNet (0.7154) and EEG Conformer (0.7400) on ISRUC, whereas the same architecture randomly initialized reaches only 0.5657/0.5003—a +0.19/+0.21 BA gap. The spectral pretext task therefore produces a representation that is discriminative off-the-shelf, not only after fine-tuning.

Label efficiency. A probe on our frozen features reaches high accuracy with far fewer downstream labels than a random encoder (Fig. 3). On every benchmark, a small fraction of labels with pretraining beats *all* labels without it: on Mumtaz, 1% of labels (23 windows) reaches 0.8990 BA, above a random encoder using 100% (0.8200); on ISRUC, 10% of sequences reaches 0.6520, above random at 100% (0.5540); similarly for HMC (0.5560 at 10% vs. 0.4920 at 100%) and Mental Arithmetic (0.5790 at 1% vs. 0.5160 at 100%). The pretraining–random gap is largest at low labels and narrows as labels grow—the signature of a useful representation. On the sequence tasks the gap emerges at 10%: below that, too few sequences exist to fit the temporal head at all (both near chance), so pretraining lowers the label threshold at which the task becomes learnable by an order of magnitude.

Table 5: **Objective ablation** at ~2,500 h, matched compute (same subset, same steps; only the loss changes). Isolates spectral band-power prediction (CF) vs. raw-signal reconstruction (MAE). Reported on the low-variance sleep benchmarks (large test sets, std ~0.005), where a small delta is resolvable; the tiny-cohort clinical tasks are too high-variance to separate the conditions.

Objective	ISRUC BA	HMC BA
MAE only (reconstruction)	?	?
CF only (spectral prediction)	?	?
CF + MAE (ours)	**0.7934**	**0.7484**

Ablations

Spectral target vs. signal reconstruction. Table 5 compares, at identical data and compute, our band-power target (CF) against pure raw-signal reconstruction (MAE) on the same backbone. *[Result pending: if CF-only ≈ full > MAE-*

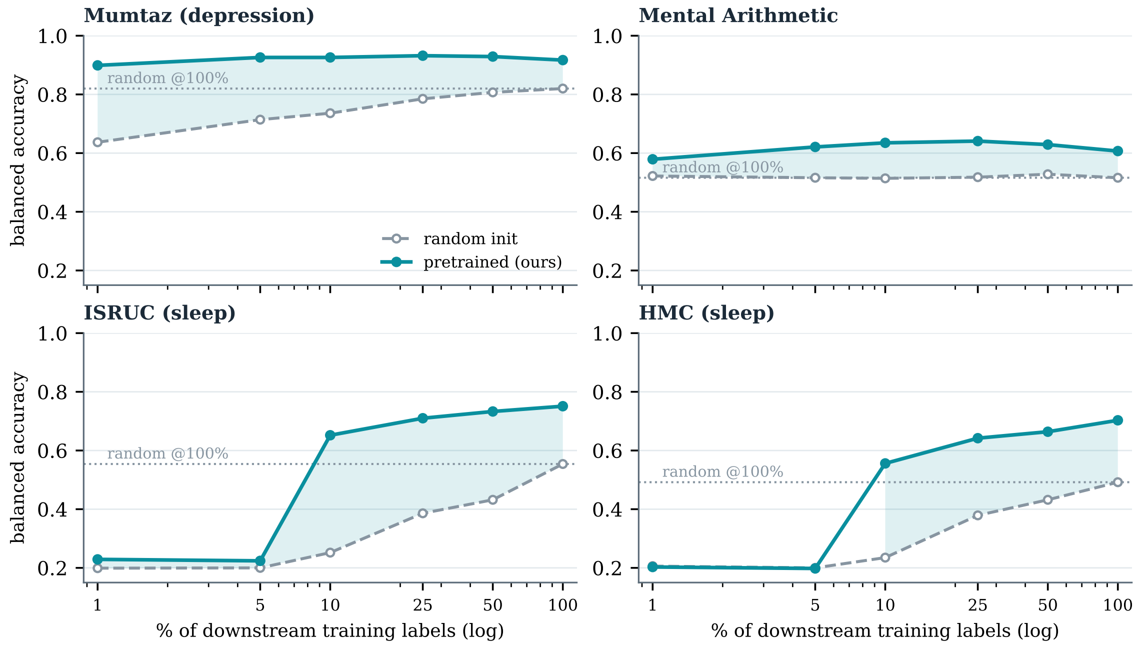

Figure 3: **Label efficiency** of the ∼2,500-hour model. Balanced accuracy of a probe trained on a fraction of downstream labels, pretrained (ours, solid) vs. random init (dashed); dotted line is random at 100% labels, shaded band is the pretrained−random gap. A small fraction of labels with pretraining exceeds all labels without it on every task; on the sequence tasks (ISRUC, HMC) the advantage appears once $\geq 10\%$ of sequences suffice to fit the head.

only, spectral prediction is the load-bearing ingredient and reconstruction is unnecessary.]

Fixed vs. learnable target. A learnable-target filterbank collapses ($\mathcal{L}_{\mathrm{cf}} \to 0$, degenerate features); the fixed target is necessary and sufficient to prevent this.

Limitations

Our model is designed for the spectral, quasi-stationary structure of clinical and passive EEG. It is *not* suited to motor imagery, whose discriminative signal is a spatially localized event-related (de)synchronization rather than a stationary band-power pattern; we exclude motor-imagery benchmarks by design rather than reporting weak numbers. Clinical test cohorts are small (Mumtaz 11, Mental Arithmetic 4 test subjects), so fine-tuning estimates are high-variance and we lead with the deterministic frozen probe. We also exclude TUH-derived abnormality detection (TUAB) from the main comparison: it overlaps at the patient level with TUEG pretraining for every method, and is off-topic for our spectral scope.

Conclusion

Predicting band power—directly, against a fixed filterbank target, with no codebook and no two-stage tokenizer—is a simple objective that is well matched to the spectral structure of clinical EEG. It reaches state-of-the-art sleep staging with a fraction of the pretraining data used by prior foundation models, is label-efficient downstream, and makes the surprising point that for spectral EEG tasks the pretext *objective*, not the data scale, is what matters. We argue this reframes what an EEG foundation model should predict.

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